

hypervisor of choice here is the Sun xVM VirtualBox.

Beware! Most distros pack in VirtualBox OSE. The OSE here stands for Open Source Edition. VirtualBox has only been partially open sourced. Some of the advanced features have not been open sourced yet and are not available in this OSE. One of these features is VRDP support, which we need for remote access to our machines. So, we have to use the free (as in free beer) but closed source version of VirtualBox.

Some of you may ask, if we have to use closed source, why not go for VMWare Server? It's free and it's made for remote access and administration. We are going to use VirtualBox because VirtualBox is faster and more stable on Linux. Additionally, part of VirtualBox is open source. More specifically, the core engine is open source; only the VRDP bit is closed source.

This brings us to remote access and administration. Unlike VMWare Server, VirtualBox is not specifically meant for servers. It's a general hypervisor. So it has nothing like VMWare's Web Interface. On the other hand, it has a very powerful command line interface called VBoxManage, which actually exposes more features than the VirtualBox GUI. So you can SSH into your server from remote terminals and configure virtual machines. And since SSH allows you to run X applications as well, you can use the GUI to configure VMs too.

What about viewing machines? Well, that too is simple. I've been ranting all along about VRDP. VRDP (VirtualBox Remote Desktop Protocol) is VirtualBox's implementation of the Microsoft Remote Desktop Protocol implemented after it had been opened up via MSDN. This feature allows you to use a RDP client like Microsoft Remote Desktop Connection (bundled with Windows) or a desktop on Linux to see a VM's Desktop.

Now about the guest operating systems...everyone acquainted with servers should know all about RHEL, right? Well, the fact of the matter is, it's extremely expensive. The CentOS Project (Community Enterprise Operating System) makes use of the free RHEL sources, replaces the RHEL branding with CentOS branding and thus produces completely free binaries, which are a 100 per cent binary compatible with the corresponding RHEL version, thus creating a free RHEL clone.

The CentOS project even publishes the security updates. So, users of CentOS are technically using RHEL, just re-branded to escape licensing issues. By the way, RHEL and CentOS are completely separate entities and are not affiliated to each other in any way.

So let's get started, shall we?

Preparations for the launch

We need to prepare for our mission. First, there are some prerequisites that you have to set up yourself. Make sure you have a Linux or Windows (preferable) client attached to your server, which can browse

the Internet. This is to make sure that the client can communicate with the server, since all Internet bound traffic will have to be NATed by the server. Ping the server if you need more reassurance. Traceroute it if you need still more. I asked for a Windows client to be able to demonstrate the interoperability of the solution, but if you use Linux, it's just as well. I'll still give you instructions, don't worry.

Then, since we were going in a different path earlier and have changed tracks now, you need to change the partitioning scheme. Create a single LVM volume out of all the unused space of your HDD and mount it under */VMspace*. That should do.

Now, let's do some work.

Installing and configuring VirtualBox CSE

Sadly, CSE stands for closed source edition. But, there's nothing we can do about it. First of all, we need to remove VirtualBox OSE if it's there. You can use Synaptic to do it.

Then, run the following command to clean up any stale dependencies:

```
$ sudo apt-get autoremove
```

Now point your browser to www.virtualbox.org/wiki/Linux_Downloads and download the VirtualBox package for Ubuntu 8.10 (remember this is the host OS we've installed on your server). Once done, install it:

```
$ sudo dpkg -i virtualbox-2.1.2_1.4-42893_Ubuntu_intrepid_i386.deb
```

After your installation, issue the following command to resolve all dependencies:

```
$ sudo apt-get install -f
```

By the way, here's one tidbit: the closed source VirtualBox is free for personal use. This may scare you because enterprise use isn't exactly personal, is it? However, take a look at the VirtualBox Licensing FAQ Point 6, and all your fears will be put to rest. Basically, this is what it says:

"Personal use is when you install the product on one or more PCs yourself and you make use of it (or even your friend, sister and grandmother). It doesn't matter whether you just use it for fun or run your multi-million Euro business with it. Also, if you install it on your work PC at some large company, this is still personal use. However, if you are an administrator and want to deploy it to the 500 desktops in your company, this would no longer qualify as personal use. Well, you could ask each of your 500 employees to install VirtualBox but don't you think we deserve some money in this case? We'd even assist you with any issue you might have."